

Reference Sheet for a Thesis with L^AT_EX2e and KOMA-Script

- All examples were tested with pdflatex.
- The package mentioned in the headings has to be included (see B.2).
- Compile three times after last change (esp. docs with references).

A. L^AT_EX Basics

A.1. Units

- Available units for length and dimensions:
bp point (typographic) mm millimeter in inch em width of M
px pixel (1/72in) cm centimeter pc pica ex height of x
- Document dependent units z\textwidth, z\linewidth, z\columnwidth, z\textheight with z a percentage value, e.g. 0.55\textwidth means 55% of the actual width of the text.
- \baselineskip minimum vertical space between the bottom of two successive lines in a paragraph.
- Amounts like \smallskipamount, \medskipamount, \bigskipamount.

A.2. Reserved Characters (see also E.2, cf. H)

\	introduces a command	\textbackslash
{ }	embraces arguments, creates logical parts	\{ \}
[]	embraces optional arguments	[]
%	comments: code after % will be ignored.	%
&	separates columns in tabular-like environments	&
#	parameter for own command declarations	#
\$	text style math mode (abbr. for \(...\))	\$
^	index/exponent only valid in math mode, e.g. a ₁ ² see E.2	

B. Preamble (before \begin{document})

B.1. Documentclass (necessary)

Use: \documentclass[opt,opt,...]{class}
Recommended classes: scrartcl, scrreprt, scrbook, scrllttr2
Non-KOMA-Script classes: beamer, koma-moderncvclassic

Common options with default	Values available (subtotal)
fontsize=11pt	10pt 12pt (e.g. 12.5pt also valid)
paper=a4, paper=portrait	a3 a5 b4 letter, landscape
parskip=no	half full
headings=big	small normal
chapterprefix=false	true
open=right (scrbook)	any (scrartcl, scrreprt) left
captions=oneline	nooneline
captions=tablebelow,figurebelow	tableabove, figureabove
toc=notlistof	listof listofnumbered
bibliography=totoc totocnumbered	nottotoc
twoside=true (scrbook)	false (scrartcl, scrreprt)
twocolumn=false	true
draft=false	true (show overfull boxes)

→ Options of document class are passed to every loaded package.
→ Set or change options later in file, e.g. \KOMAoptions[twoside=true]

B.2. Loading Packages

\usepackage[options]{package}
\PassOptionsToPackage[options]{package}

B.3. Encoding Settings

```
\usepackage[utf8]{inputenc} % most IDEs use UTF8
\usepackage[T1]{fontenc} % most fonts needs T1
```

B.4. Language Settings with babel

Load: \usepackage[ngerman, main=english]{babel}
Use: \selectlanguage{language} \foreignlanguage{language}{text}

```
\documentclass[italian]{scrbook} % global option
\usepackage[british,main=italian]{babel} % package option
\usepackage{csquotes} % package csquotes knows italian
```

C. Layout

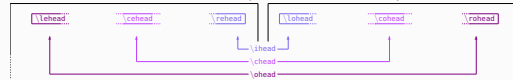
C.1. Changing Page Layout with geometry

- Let KOMA-Script know of geometry by option usegeometry=true.

```
\usepackage[left=2cm, right=2, top=3cm, bottom=4cm,
bindingoffset=1cm, includeheadfoot]{geometry}
```

- Auto-completion determines unspecified dimensions (under or over specified as well), here width and height of text (see I.9).
- Other options: paper=a4paper, landscape|portrait, includehead, includefoot, includeheadfoot, twocolumn
- Changing page layout mid document: \newgeometry{opt, opt, ...}

C.2. Header and Footer of Page (aka running heading)



```
% delete default settings and define your own
\usepackage[automark]{scrlayer-scrpage}
\clearpaairofpagestyles
\ohead[]{\headmark} \ofoot[\pagemark]{\pagemark}
```

```
% Variant for a thesis with horizontal rules at head and foot
\usepackage[headsepline=0.005pt:,footsepline=0.005pt:,
plainfootsepline,automark]{scrlayer-scrpage}
\clearpaairofpagestyles
\ohead[]{\headmark} \ofoot[\pagemark]{\pagemark}
\ModifyLayer[addvoffset=-.6ex]{scrheadings.foot.above.line}
\ModifyLayer[addvoffset=-.6ex]{plain.scrheadings.foot.above.line}
\setkomafont{pageheadfoot}{\small}
```

C.3. Linespread with setspace

Load: \usepackage[onehalfspacing]{setspace} for 1.5 line spacing.

D. Document Structure (see also L)

D.1. Start Document

```
\begin{document} Complete document contents. \end{document}
```

D.2. Title

simple title: \author{text} \title{text} \date{\today} \maketitle
title page self designed: \begin{titlepage} text \end{titlepage}

D.3. Table of Contents, List of Figures (for other List of see E.8 & G)

\tableofcontents \listoftables \listoffigures
KOMAoption toc=listof (see B.1) generates entries for TOC.

D.4. Headings

```
\part{title} \chapter{title}
\section{title} \subsection{title} \subsubsection{title}
\paragraph{title} \subparagraph{title}
```

- \chapter only valid in documentclass scrbook and scrreprt
- Use * variants for headings without numbering, no change in counter and no entry in table of contents.
- Use the optional parameter for short titles in headings and table of contents, e.g. \section[short title]{title}
- Use \addpart, \addchap or \addsec for unnumbered headings, but with running heading and entry in table of contents. The * variants delete the running heading.
- Layout of paragraph and subparagraph similar to other headings:
\RedeclareSectionCommands[afterskip=1sp]{paragraph, subparagraph}
\setcounter{secnumdepth}{\subparagraphnumdepth}
\setcounter{tocdepth}{\subparagraphtocdepth}

D.5. Justification

Environment	Declaration	Other
\begin{center}	\centering	text \par\vfill text
\begin{flushleft}	\raggedright	text \hfill text
\begin{flushright}	\raggedleft	\raggedbottom, \flushbottom

D.6. Lists

```
\begin{itemize} with bullets \item or \item[symbol]
\begin{enumerate} with numbers \item
\begin{description} with bold words \item[word]
\begin{labeling}[separator]{labelinglabel} \item[word]
```

```
\begin{enumerate}
\item First item
\item Second item\label{it:second} % see References
\end{enumerate}
```

D.7. Enhanced Lists with enumitem

Load: \usepackage{enumitem}

Example (for enumerate):

```
\setlist[enumerate,1]{label=\Alph*}} A) one
\setlist[enumerate,2]{label=\alpha*}} a) one
\setlist[enumerate,3]{label=\roman*}} b) two
\setlist[enumerate,4]{label=\arabic*}} B) two
```

Example (for legal list):

```
\newlist{legal}{enumerate}{10} 1. one
\setlist[legal]{label=\arabic*,.noitemsep} 1.1. two
Use: \begin{legal} \item ... \end{legal} 1.1.1. three
1.1.2. strawberry
```

D.8. Separate Files

- After preamble within the text place: \include{file} Text starts and ends on a new page. file has to be in the same directory as the master file. Otherwise specify a path: \include{path/file}
- In preamble place: \includeonly{file1,file2} to run only these files.
- Use \input{file} includes a file without starting/ending on a new page (\includeonly not valid).

E. Text

E.1. Paragraphs (≈ "new idea in content")

Paragraphs are separated by an empty line in the code or by \par.
A \\ produces a new line – use sparingly, seldom needed outside tabulars.
Correct Overfull Box Warnings with more than 4pt (look into log file).

E.2. Text Symbols/Characters (see also A.2)

A lot of diacritic symbols can be typed directly, e.g. è é ê ñ ç

```
§ \S _ \textunderscore{} ~ \textasciitilde{}
^ \textasciicircum{} ... \ldots | \textbar
```

Other symbols need packages, e.g. € \texteuro (textcomp)

E.3. Fonts

Command	Declaration	Effect
\textrm{text}	\rmfamily text	Roman family
\textsf{text}	\sffamily text	Sans serif family
\texttt{text}	\ttfamily text	Typewriter family
\textmd{text}	\mdseries text	Medium series
\textbf{text}	\bfseries text	Bold series
\textup{text}	\upshape text	Upright shape
\textit{text}	\itshape text	<i>Italic shape</i>
\textsl{text}	\slshape text	<i>Slanted shape</i>
\textsc{text}	\scshape text	SMALL CAPS SHAPE

More general commands:
\emph{text} \em text *Emphasized*
\textnormal{text} \normalfont text Document font

Example: \setkomafont{section}{\scshape}

E.4. Font Size

Font size is relative to the base font size, specified in the document class.

```
\tiny tiny \Large Large
\scriptsize scriptsize \LARGE LARGE
\footnotesize footnotesize
\small small \huge huge
\normalsize normalsize
\large large \Huge Huge
```

Use: \small text or \huge text\par to limit the size change.

Example: \setkomafont{pageheadfoot}{\small}

E.5. Colors with xcolor

```
\usepackage{xcolor}
\definecolor{DarkBlue}{RGB}{0, 115, 207}
\colorlet{col.section}{DarkBlue}
\textcolor{red}{text in red} or {\color{red}text}
\colorbox{gray!25}{\color{gray} faded by 25\%}
```

Predefined colors:

white gray black red green blue cyan magenta yellow

Fade a color with color!value between 0 and 100

Headings in color: \setkomafont{disposition}{\color{color}}

E.6. Footnotes

\footnote{text}	Print footnote marker in text and footnote at bottom of page
\footnotemark	Print footnote marker in text (e.g. within tabular or caption)
\footnotetext{text}	Print footnote at bottom of page

E.7. References with hyperref (loads url implicitly)

\autocite{citekey}	Cite a bibliographic reference (package biblatex)
\label{marker}	Set a marker for cross reference, often if the form \label{sec:item} or \label{fig:diag1}
\autoref{marker}	Give type name and number of marker
\autopageref{marker}	Give abbreviation of "page" and page number of marker
\url{url}	Print clickable web page
\href[options]{url}{text}	Print clickable link
\hyperref[marker]{text}	Print clickable reference

Style: \urlstyle{xx} with xx a style like "tt", "rm", "sf" or "same".

Names for autoref (package babel):

```
\renewcaptionname{language}{\typenameautorefname}{text},
e.g. \renewcaptionname{english}{\subsectionautorefname}{section}
```

E.8. Acronyms with acro

```
\usepackage{acro,hyperref,longtable,tabu} \next 5 to praeambel
\acsetup{list-style=longtabu,list-heading=addchap}
\DeclareAcroListStyle{longtabu}{table}{table=longtabu,
table-spec=@{}{\bfseries}\X@{}}
\DeclareAcronym{ecm}{short=EM,long=Electro Machining}
...
\ac{EM} or \Ac{EM} for capitalized first letter
\printacronyms
```

F. Figures & Tables (floating environments)

F.1. Figures with graphicx

Load: \usepackage{graphicx}
Use: \includegraphics[opt]{file} (png, jpg, pdf)

With 'figure' the environment to place a graphic is meant. The figure caption is printed where the caption command is placed in the input. Extra vertical space is controlled by the KOMAoption captions (see B.1).
Use: \begin{figure}[pos] ..\caption{..}\label{fig:x} \end{figure}
Parameter: pos is a suggestion for placing, it can be ignored by T_EX.
Possible values are combinations of t (top), h (here), b (bottom), ! (try harder), p (separate page).
Hint: Define a path to the graphic files (no blanks in folder names; no special characters in file names) \graphicspath{ \folder/}{\folder/}... }

```
\graphicspath{ {img/} } %subfolder for images; set in preamble
\begin{figure}\centering
\includegraphics[width=.8\columnwidth]{pic.jpg}
\caption[Short title]{Long title}\label{fig:ff}
\end{figure}
```

- Numbering throughout the whole document (scrbook) with package chngcntr: \counterwithout{figure}{chapter} (same for table)
- Figure name: \renewcaptionname{language}{\figurename}{text}
\renewcaptionname{language}{\figureautorefname}{text}

```

Load: \usepackage{subcaption}
Use: \begin{subfigure}[pos]{width} ... \end{subfigure}

\begin{figure}[ht] \centering
\begin{subfigure}[t]{0.5\textwidth}
\centering \includegraphics[height=1.2in]{figure-a}
\caption{Subcaption 1}\label{fig:SubFig1}\end{subfigure}
\begin{subfigure}[t]{0.5\textwidth}
\centering \includegraphics[height=1.2in]{figure-b}
\caption{Subcaption 2}\label{fig:SubFig2}\end{subfigure}
\caption{Caption of complete figure}\label{fig:Fig1}
\end{figure}

```

With 'table' the environment to place aligned material is meant. The table caption is printed where the caption command is placed in the input. For positioning options see [F.1](#).

Normal: `\parbox[pos][height][contentpos]{width}{text}` or
`\begin{minipage}[pos][height][contentpos]{width}text\end{minipage}`
 Lift Text: `\raisebox[lift][height][depth]{text}`
 Framed Box: `\fbox{text}` or `\framebox[width][pos]{text}`
 Colored Box: `\xcolor{color}\colorbox{backgroundcolor}{text}`
 Framed colored Box: `\fcolorbox{bordercolor}{backgroundcolor}{text}`
 Resize (graphicx): `\scalebox{10}{giant}`
 Lengths: `\setlength{fboxsep}{unit}`, `\setlength{fboxrule}{unit}`

$\arccos$ $\arcsin$ $\arctan$ $\arg$ $\cos$ $\cosh$ $\cot$ $\coth$ $\csc$ $\deg$ $\det$
 $\dim$ $\exp$ $\gcd$ $\hom$ $\inf$ $\ker$ $\lg$ $\lim$ $\liminf$ $\limsup$
 $\ln$ $\log$ $\max$ $\min$ $\Pr$ $\sec$ $\sin$ $\sinh$ $\sup$ $\tan$ $\tanh$

For other functions use (package `amsmath`): name , e.g. $\operatorname{operatorname{arcsinh}}$ (see also [J.2](#)).

<pre>\begin{align} y & \mathrel{\mathop:}= d \\ y & \mathrel{\mathop:}= cx+d\nonumber \\ y & \mathrel{\mathop:}= bx^2+cx+d \label{eq:key} \end{align}</pre>	$y = d \tag{1}$ $y = cx + d$ $y = bx^2 + cx + d \tag{2}$
<pre>\begin{align*} y & \mathrel{\mathop:}= d & z & \mathrel{\mathop:}= 1 \\ y(x) & \mathrel{\mathop:}= cx+d & z & \mathrel{\mathop:}= x+1 \\ y_{12} & \mathrel{\mathop:}= bx^2+cx & z & \mathrel{\mathop:}= x^2+x+1 \end{align*}</pre>	$y = d \qquad z = 1$ $y(x) = cx + d \qquad z = x + 1$ $y_{12} = bx^2 + cx \qquad z = x^2 + x + 1$

$\mapsto$	<code>\mapsto</code>	$\rightsquigarrow$	<code>\leadsto</code>
$\rightarrow$	<code>\rightarrow</code>	$\Rightarrow$	<code>\Rightarrow</code>
$\longrightarrow$	<code>\longrightarrow</code>	$\Longrightarrow$	<code>\Longrightarrow</code>
$\leftarrow$	<code>\leftarrow</code>	$\Leftarrow$	<code>\Leftarrow</code>
$\longleftarrow$	<code>\longleftarrow</code>	$\Longleftarrow$	<code>\Longleftarrow</code>
$\uparrow$	<code>\uparrow</code>	$\Uparrow$	<code>\Uparrow</code>
$\downarrow$	<code>\downarrow</code>	$\Downarrow$	<code>\Downarrow</code>
$\leftrightarrow$	<code>\leftrightarrow</code>	$\Leftrightarrow$	<code>\Leftrightarrow</code>
$\leftrightsquigarrow$	<code>\leftrightsquigarrow</code>	$\rightsquigarrow$	<code>\rightsquigarrow</code>
$\leftrightharpoons$	<code>\leftrightharpoons</code>	$\rightleftharpoons$	<code>\rightleftharpoons</code>
$\leftrightsquigarrow$	<code>\leftrightsquigarrow</code>	$\rightleftharpoons$	<code>\rightleftharpoons</code>

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Head	Head	Head											
Data	first	Row											
Data	second	Row											
2. Method Some text, see Table 1.1 for aligned material. L'Hopital's rule: (2.1) $\lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$ 2	2. Method More text. A "quote" and a „Zitat“. 3	A. Appendix A BOX Figure A.1.: Example of a caption beside the figure 4	Bibliography Wikipedia. <i>TeX - Wikipedia, The Free Encyclopedia</i>. https://en.wikipedia.org/wiki/TeX, last referenced 24-June-2017. 5	License Reference Sheet, Version 1.2, August 2017 This Cheat Sheet is free under the terms and conditions of the L^AT_EX Project Public License Version 1.3c. Authors: Marion Lammarch, University of Heidelberg, and Elke Schubert, Stutenrose Credit for the layout and content ideas goes to (a) Emanuel Regnath, "L^AT_EX Cheat Sheet" L^AT_EX 2_ε 2013 http://www.latexte.de/downloads/LaTeX_CheatSheet.pdf (b) Tanno Schwidh, DHBW Mosbach, "L^AT_EX Cheat Sheet German", Ed. 2014 http://mirror.ctan.org/tex-archive/info/latexcheat/latexcheat-de/ The logo with the T_EX lion is sketched by Dwayne Hilby; https://mirror.ctan.org/1306/t1306; the graphic of a header of a page is from the KOMA-Script documentation, file scrlayer-scrpage.tex; the page layout image is done by using the package layout and the command \layout and the red lines and red text added with a graphics program. Acknowledgements We would like to thank Markus Kohn, Ulrike Fischer, Martin Sievers, and Herbert Voß for their useful hints and suggestions. Not naming all the female and male students which with their intrigued questions helped to improve this reference sheet.									